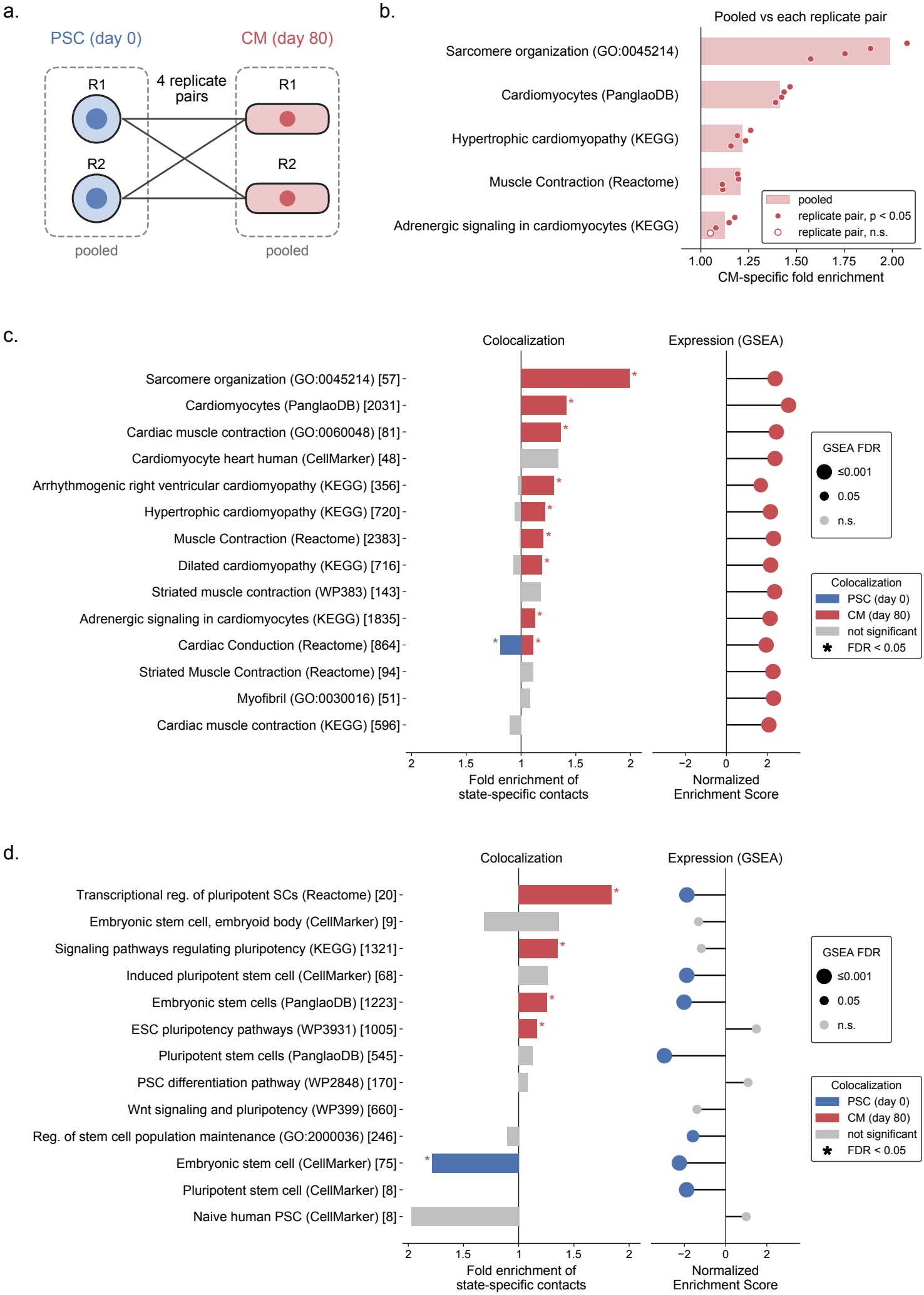
a.
PSC (day 0)
CM (day 80)
R1
R2
4 replicate pairs
R1
R2
pooled
pooled
b.
Pooled vs each replicate pair
Sarcomere organization (GO:0045214)
Cardiomyocytes (PanglaoDB)
Hypertrophic cardiomyopathy (KEGG)
Muscle Contraction (Reactome)
Adrenergic signaling in cardiomyocytes (KEGG)
pooled
replicate pair, p < 0.05
replicate pair, n.s.
1.00
1.25
1.50
1.75
2.00
CM-specific fold enrichment
c.
Colocalization
Expression (GSEA)
Sarcomere organization (GO:0045214) [57]
Cardiomyocytes (PanglaoDB) [2031]
Cardiac muscle contraction (GO:0060048) [81]
Cardiomyocyte heart human (CellMarker) [48]
Arrhythmogenic right ventricular cardiomyopathy (KEGG) [356]
Hypertrophic cardiomyopathy (KEGG) [720]
Muscle Contraction (Reactome) [2383]
Dilated cardiomyopathy (KEGG) [716]
Striated muscle contraction (WP383) [143]
Adrenergic signaling in cardiomyocytes (KEGG) [1835]
Cardiac Conduction (Reactome) [864]
Striated Muscle Contraction (Reactome) [94]
Myofibril (GO:0030016) [51]
Cardiac muscle contraction (KEGG) [596]
2
1.5
1
1.5
2
Fold enrichment of state-specific contacts
−2
0
2
Normalized Enrichment Score
GSEA FDR
≤0.001
0.05
n.s.
Colocalization
PSC (day 0)
CM (day 80)
not significant
FDR < 0.05
d.
Colocalization
Expression (GSEA)
Transcriptional reg. of pluripotent SCs (Reactome) [20]
Embryonic stem cell, embryoid body (CellMarker) [9]
Signaling pathways regulating pluripotency (KEGG) [1321]
Induced pluripotent stem cell (CellMarker) [68]
Embryonic stem cells (PanglaoDB) [1223]
ESC pluripotency pathways (WP3931) [1005]
Pluripotent stem cells (PanglaoDB) [545]
PSC differentiation pathway (WP2848) [170]
Wnt signaling and pluripotency (WP399) [660]
Reg. of stem cell population maintenance (GO:2000036) [246]
Embryonic stem cell (CellMarker) [75]
Pluripotent stem cell (CellMarker) [8]
Naive human PSC (CellMarker) [8]
2
1.5
1
1.5
2
Fold enrichment of state-specific contacts
−2
0
2
Normalized Enrichment Score
GSEA FDR
≤0.001
0.05
n.s.
Colocalization
PSC (day 0)
CM (day 80)
not significant
FDR < 0.05